

* Classical Pure Spincer String:

$$S_c = \int d\tau d\sigma \left\{ p_m \dot{x}^m + \omega_a \nabla \lambda^a + \hat{\omega}_a \nabla \hat{\lambda}^a - \frac{1}{2} L(\dot{x} - \partial_\sigma x) \lambda + K \nabla_\sigma \lambda + e \tau - \frac{1}{2} \hat{L}(\dot{x} - \partial_\sigma x) \hat{\lambda} + \hat{k} \nabla_\sigma \hat{\lambda} + e \bar{\tau} \right\}$$

• Can, be, conformal symmetry, shift:

$$\lambda^\alpha \mapsto \lambda^\alpha + \frac{(\dot{x} - \partial_\sigma x) \bar{\lambda}}{\lambda \bar{\lambda}} e$$

$$K_\alpha \mapsto K_\alpha + \frac{N_m (\gamma^m \bar{\lambda})_\alpha + W \lambda_\alpha}{\lambda \bar{\lambda}} e$$

$$T = -\frac{1}{2} (\dot{x} + \partial_\sigma x)^2 + \omega_a \nabla_\sigma \lambda^a$$

* Can define the BRST operator of the theory via the exact action $\int Q^\mu \chi$

• $Q\mathcal{U} = \partial V$ (for superfields/wave functions)

$$V = \lambda^\alpha A_\alpha(x, \theta), \quad A_\alpha(x, \theta): \text{Spincer superfield}$$

$$\mathcal{U} = \partial_\theta^\alpha A_\alpha(x, \theta) + \pi^m A_m(x, \theta) + d_\alpha W^\alpha(x, \theta) + N_{mn} F^{mn}(x, \theta) + \frac{1}{2} N^{mn} (\gamma_{mn} \lambda)$$

* Gauge fix: d_α : Define the primary constraints which fix Q

$$Q = \int d^2 z \lambda^\alpha d_\alpha = \int d^2 z \lambda^\alpha \left(p_\alpha - \frac{1}{2} \partial x^m \gamma_{m\alpha} - \frac{1}{2} (\partial_\sigma \gamma^\sigma_\alpha) \tau_{mb} \right)_a$$

$$A_m = \frac{1}{8} D_\alpha \gamma_{m\alpha}^{\alpha\beta} A_\beta$$

$$W^\beta = \frac{1}{10} \gamma_m^{\alpha\beta} (D_\alpha A^m - \partial^m A_\alpha) \quad \hookrightarrow p_\alpha: \text{independent world sheet web}$$

$$F_{mn} = \frac{1}{8} D_\alpha (\gamma_{mn})^\alpha_\beta W^\beta$$

* The QPE of $d_\alpha(\chi) d_\beta(z)$:

$$d_\alpha(\chi) d_\beta(z) \sim \frac{\gamma_{\alpha\beta}^m \pi_m}{y-z} = \frac{\gamma_{\alpha\beta}^m (\partial x_m + \frac{1}{2} \partial_\sigma \gamma_{m\sigma})}{(y-z)}$$

Q : Manifestly Super-poincare invariant

V : Vertex operator of ghost $\neq 1$

* Q , to define physical fields, must satisfy

$$\psi_{\text{physical}} = Q\psi \quad \text{set} \quad Q\psi = 0$$

$$- \text{i.e. } \psi \in H(M^{10|4})$$

$$\text{or } Q^2 = 0$$

↳ Proof:

$$Q^2 = \int d^2z \left(\lambda^\alpha \left(p_\alpha - \frac{1}{2} \gamma_{\alpha\beta}^\mu \theta^\beta \partial X_\mu - \frac{1}{8} \gamma_{\alpha\beta}^\mu (\dot{X}_\mu) \gamma_{\beta\gamma} \theta^\gamma \partial \theta^\delta \right) \right)$$

$$= \int \left(\lambda^\beta p_\beta \lambda^\alpha p_\alpha - \frac{1}{2} \lambda^\alpha p_\alpha \lambda^\beta \gamma_{\beta\gamma}^\mu \theta^\gamma \partial X_\mu - \frac{1}{8} \lambda^\alpha p_\alpha \gamma_{\beta\gamma}^\mu (\dot{X}_\mu) \gamma_{\beta\gamma} \theta^\gamma \partial \theta^\delta \right. \\ \left. - \frac{1}{2} \lambda^\alpha \gamma_{\alpha\beta}^\mu \theta^\beta \partial X_\mu \lambda^\gamma p_\gamma + \frac{1}{4} (\lambda^\alpha \gamma_{\alpha\beta}^\mu \theta^\beta \partial X_\mu) (\lambda^\gamma \gamma_{\gamma\delta}^\mu \theta^\delta \partial X_\mu) \right. \\ \left. + \frac{1}{16} \lambda^\alpha \gamma_{\alpha\beta}^\mu \theta^\beta \partial X_\mu \lambda^\gamma \gamma_{\gamma\delta}^\mu (\dot{X}_\mu) \gamma_{\beta\gamma} \theta^\gamma \partial \theta^\delta + \right.$$

$$\text{on shell } \partial p_\alpha = 0$$

$$(a) : \lambda^\alpha p_\alpha \lambda^\beta \gamma_{\beta\gamma}^\mu \theta^\gamma \partial X_\mu + \lambda^\beta \gamma_{\beta\gamma}^\mu \theta^\gamma \partial X_\mu \lambda^\alpha p_\alpha \\ = 2 \lambda^\alpha \lambda^\beta p_\alpha \varepsilon_{\alpha\gamma} \varepsilon^{\gamma\delta} \gamma_{\beta\delta}^\mu \theta^\delta \partial X_\mu \\ = 2 \lambda^\alpha \lambda^\beta p_\alpha (-2) (\delta_\alpha^\gamma \delta_\beta^\delta - \delta_\alpha^\delta \delta_\beta^\gamma) \gamma_{\beta\delta}^\mu \theta^\delta \partial X_\mu \\ = -4 \lambda^\alpha \lambda^\beta p_\alpha \gamma_{\beta\gamma}^\mu \theta^\gamma \partial X_\mu + 4 \lambda^\alpha \lambda^\beta p_\beta \gamma_{\beta\gamma}^\mu \theta^\gamma \partial X_\mu \\ = -4 \lambda^\alpha \theta^\alpha \lambda^\beta \gamma_{\beta\gamma}^\mu p_\gamma \partial X_\mu + 0$$

$$(b) \lambda^\alpha p_\alpha \gamma_{\beta\gamma}^\mu (\dot{X}_\mu) \gamma_{\beta\gamma} \theta^\gamma \partial \theta^\delta$$

* Since $d\alpha$ define constraints they should have $\{d_\alpha, d_\beta\} \sim \gamma_{\alpha\beta}$

Will show this:

$$\text{Start w. GS: } P_\mu \dot{X}^\mu + p_\alpha \dot{\theta}^\alpha - \frac{1}{2} e p^2$$

$$\{X^\mu, P_\mu\} = \delta_\mu^\mu \quad \{\theta^\alpha, p_\alpha\} = \delta_\alpha^\alpha$$

$$d_\alpha = p_\alpha - \frac{1}{2} (\bar{\theta} \gamma^\mu)_{\alpha} \dot{X}_\mu$$

$$\{d_\alpha, d_\beta\} = \{p_\alpha - \frac{1}{2} (\bar{\theta} \gamma^\mu)_{\alpha} \dot{X}_\mu, p_\beta - \frac{1}{2} (\bar{\theta} \gamma^\mu)_{\beta} \dot{X}_\mu\} \\ = -\frac{1}{2} \{p_\alpha, \bar{\theta}^\gamma \gamma_{\gamma\beta}^\mu \dot{X}_\mu\} - \frac{1}{2} \{\bar{\theta}^\gamma \gamma_{\gamma\alpha}^\mu \dot{X}_\mu, p_\beta\} \\ = \frac{1}{2} (-\gamma_{\gamma\beta}^\mu \dot{X}_\mu \delta_\alpha^\gamma + \gamma_{\gamma\alpha}^\mu \dot{X}_\mu \delta_\beta^\gamma) \\ = \frac{1}{2} (\gamma_{\alpha\beta}^\mu \dot{X}_\mu + \gamma_{\beta\alpha}^\mu \dot{X}_\mu) \\ = -\gamma_{\alpha\beta}^\mu \dot{X}_\mu$$

$\therefore d_\alpha \sim$ supersymmetry generators

↳ Here we can set up the Pure spinor system

$$\{\theta^\alpha, p_\beta\} = -\delta_\beta^\alpha \delta(\sigma - \sigma')$$

$$\{\omega_\alpha, \lambda^\beta\} = \delta_\beta^\alpha \delta(\sigma - \sigma')$$

$$\begin{aligned}
\{d_\alpha, d_\beta\} &= \{p_\alpha - \frac{1}{2} \gamma_{\alpha\beta}^m \theta^\beta \partial x_m - \frac{1}{8} \gamma_{\alpha\beta}^m (\gamma_m)_{\gamma\delta} \theta^\gamma \partial \theta^\delta\} \\
&\quad \alpha \leftrightarrow \beta \\
&= \{p_\alpha, -\frac{1}{2} \gamma_{\beta\gamma}^m \theta^\gamma \partial x_m\} - \frac{1}{8} \{p_\alpha, (\gamma_{\alpha\epsilon}^m) (\gamma_m)_{\gamma\delta} \theta^\epsilon \theta^\gamma \partial \theta^\delta\} \\
&\quad \{ -\frac{1}{2} \gamma_{\beta\gamma}^m \theta^\gamma \partial x_m, p_\beta \} - \{ \leftarrow \rightarrow \} \\
&= -\frac{1}{2} \gamma_{\alpha\gamma}^m \partial x_m \delta_\beta^\gamma + \frac{1}{2} \gamma_{\beta\gamma}^m \partial x_m \delta_\alpha^\gamma \\
&\quad - \frac{1}{8} (\gamma^m)_{\alpha\epsilon} (\gamma_m)_{\gamma\delta} \{p_\alpha, \theta^\epsilon \theta^\gamma \partial \theta^\delta\} + \dots \\
&= -\frac{1}{2} \gamma_{\alpha\beta}^m \partial x_m + \frac{1}{2} \gamma_{\beta\alpha}^m \partial x_m - \frac{1}{8} (\gamma^m)_{\alpha\epsilon} \dots
\end{aligned}$$

$$\begin{aligned}
& * \{p_\alpha, \theta^\epsilon \theta^\gamma \partial \theta^\delta\} \\
&= p_\alpha \theta^\epsilon \theta^\gamma \partial \theta^\delta + \theta^\epsilon \theta^\gamma \partial \theta^\delta p_\alpha \\
&= p_\alpha \theta^\epsilon \theta^\gamma \partial \theta^\delta - \theta^\epsilon (-\delta_\alpha^\gamma - p_\alpha \theta^\gamma) \partial \theta^\delta \\
&= p_\alpha \theta^\epsilon \theta^\gamma \partial \theta^\delta + \theta^\epsilon \partial \theta^\delta \delta_\alpha^\gamma + \theta^\epsilon p_\alpha \theta^\gamma \partial \theta^\delta \\
&= p_\alpha \theta^\epsilon \theta^\gamma \partial \theta^\delta + \theta^\epsilon \partial \theta^\delta \delta_\alpha^\gamma - p_\alpha \theta^\epsilon \theta^\gamma \partial \theta^\delta \\
&\quad - \delta_\alpha^\epsilon \theta^\gamma \partial \theta^\delta
\end{aligned}$$

$$\begin{aligned}
\therefore \{p_\alpha, \theta^\epsilon \theta^\gamma \partial \theta^\delta\} &= \theta^\epsilon \partial \theta^\delta \delta_\alpha^\gamma - \delta_\alpha^\epsilon \theta^\gamma \partial \theta^\delta \\
&\quad + \frac{1}{8} (\gamma^m)_{\beta\epsilon} (\gamma_m)_{\gamma\delta} (\theta^\epsilon \partial \theta^\delta \delta_\alpha^\gamma - \delta_\alpha^\epsilon \theta^\gamma \partial \theta^\delta) \\
&\quad + \frac{1}{8} (\gamma^m)_{\beta\epsilon} (\gamma_m)_{\alpha\delta} \theta^\epsilon \partial \theta^\delta = (\gamma^m)_{\beta\alpha} (\gamma_m)_{\gamma\delta} \theta^\epsilon \partial \theta^\delta \\
&= \frac{1}{8} (\gamma^m \theta)_{\beta} (\gamma_m \partial \theta)_{\alpha} - \frac{1}{8} (\gamma^m)_{\beta\alpha} (\theta \gamma_m \partial \theta)
\end{aligned}$$

$$\{d_\alpha, d_\beta\} = -\gamma_{\alpha\beta}^m \pi_m \quad \left| \quad \pi^m = \partial x^m + \frac{1}{2} \theta \gamma^m \partial \theta \right.$$

Q^2 : defined as $\{Q, Q\}$ in the CFT sense

$$\begin{aligned}
\therefore \{Q, Q\} &= Q^2 = \oint d^2 z \lambda^\alpha \lambda^\beta \{d_\alpha, d_\beta\} \\
&= -\oint d^2 z \lambda^\alpha \lambda^\beta \gamma_{\alpha\beta}^m \pi_m = -\oint d^2 z \lambda \gamma^m \lambda \pi_m
\end{aligned}$$

* For $D=10$: Pure spinors $\lambda \gamma^m \lambda = 0$
 Breaks $Sp(10) \rightarrow SU(5)$ reps

$$\begin{aligned}
& * \text{Explicit form of } \{d_\alpha, d_\beta\} = -\gamma_{\alpha\beta}^m \pi_m \\
&= -\gamma_{\alpha\beta}^m (\partial x_m + \frac{1}{2} \theta \gamma_m \partial \theta)
\end{aligned}$$

$$\{\theta^\alpha, p_\beta\} = \delta_\beta^\alpha \delta(\sigma - \sigma')$$

$$\text{E.o.m of } p_\alpha \quad \bar{\partial} p_\alpha = 0$$

$$\begin{aligned}
d_\alpha &= p_\alpha - \frac{1}{2} (\gamma^m \theta)_\alpha \pi_m \\
&= p_\alpha - \frac{1}{2} (\gamma^m \theta)_\alpha (\partial x_m + \frac{1}{2} \theta \gamma_m \partial \theta)
\end{aligned}$$

$$= p_\alpha - \frac{1}{2} \gamma_{\alpha\beta}^m \theta^\beta \partial_m - \frac{1}{8} \gamma_{\alpha\epsilon}^m \theta^\epsilon (\gamma_m)_\gamma \delta \theta^\gamma \partial \theta^\delta$$

Since $\theta \gamma_m \partial \theta^\delta = \theta^\gamma (\gamma_m)_\gamma \delta \partial \theta^\delta$

$$= \frac{1}{2} (\theta^\gamma (\gamma_m)_\gamma \delta \partial \theta^\delta + \theta^\delta (\gamma_m)_\delta \gamma \partial \theta^\gamma)$$

$$= \frac{1}{2} (\theta^\gamma (\gamma_m)_\gamma \delta \partial \theta^\delta + \theta^\delta (\gamma_m)_\delta \gamma \partial \theta^\gamma)$$

$$= \frac{1}{2} (\gamma_m)_\gamma \delta (\theta^\gamma \partial \theta^\delta - \theta^\delta \partial \theta^\gamma)$$

$$= (\gamma_m)_\gamma \delta (\theta^{\gamma\gamma} \partial \theta^\delta)$$

* Go back to $M(10|15)$: $\Pi_m = \dot{x}_m + \frac{1}{2} \theta \gamma_m \dot{\theta}$

$$\therefore d\alpha = p_\alpha - \frac{1}{2} \gamma_{\alpha\beta}^m \theta^\beta \Pi_m - \frac{1}{8} \gamma_{\alpha\beta}^m (\gamma_m)_\gamma \delta \theta^\gamma \partial \theta^\delta$$

$$\{\theta^\alpha, p_\beta\} = \delta_\beta^\alpha \delta(\sigma - \sigma')$$

$$\{d\alpha, d\beta\} = \{p_\alpha, (-\frac{1}{2} \gamma_{\beta\gamma}^m \theta^\gamma \Pi_m)\}$$

$$+ \{p_\alpha, (-\frac{1}{8} \gamma_{\beta\epsilon}^m (\gamma_m)_\gamma \delta \theta^\gamma \partial \theta^\delta)\}$$

$$= -\frac{1}{2} (-\delta_\alpha^\gamma \delta(\sigma - \sigma')) \gamma_{\beta\gamma}^m \Pi_m$$

$$\textcircled{a} -\frac{1}{8} (\gamma_{\beta\epsilon}^m) (\gamma_m)_\gamma \delta \{p_\alpha, \theta^\epsilon \partial \theta^\gamma \partial \theta^\delta\} \quad \hookrightarrow \alpha = \gamma \beta$$

E.O.M of $\bar{P} = 0$:

$$\textcircled{a} \{p_\alpha, \theta^\epsilon \partial \theta^\gamma \partial \theta^\delta\} = (p_\alpha \theta^\epsilon \partial \theta^\gamma \partial \theta^\delta) + \theta^\epsilon \partial \theta^\gamma \partial \theta^\delta p_\alpha$$

$$= () + \theta^\epsilon \partial \theta^\gamma (-p_\alpha \partial \theta^\delta + \delta_\alpha^\gamma \partial_\tau \delta(\sigma - \sigma'))$$

$$= () - \theta^\epsilon \partial \theta^\gamma p_\alpha \partial \theta^\delta + \theta^\epsilon \partial \theta^\gamma \delta_\alpha^\gamma \partial_\tau \delta(\sigma - \sigma')$$

$$= () + \theta^\epsilon p_\alpha \partial \theta^\gamma \partial \theta^\delta - \theta^\epsilon \delta_\alpha^\gamma \delta(\sigma - \sigma') \partial \theta^\delta + \theta^\epsilon \partial \theta^\gamma \delta_\alpha^\gamma \partial_\tau \delta(\sigma - \sigma')$$

$$= () - p_\alpha \theta^\epsilon \partial \theta^\gamma \partial \theta^\delta + \delta_\alpha^\epsilon \delta(\sigma - \sigma') \partial \theta^\gamma \partial \theta^\delta$$

$$- \theta^\epsilon \delta_\alpha^\gamma \delta(\sigma - \sigma') \partial \theta^\delta + \theta^\epsilon \partial \theta^\gamma \delta_\alpha^\gamma \partial_\tau \delta(\sigma - \sigma')$$

$$= (\delta_\alpha^\epsilon \partial \theta^\gamma \partial \theta^\delta - \delta_\alpha^\gamma \partial \theta^\epsilon \partial \theta^\delta + \theta^\epsilon \partial \theta^\gamma \delta_\alpha^\gamma \partial_\tau) \delta(\sigma - \sigma')$$

$$\therefore (-\frac{1}{8} (\gamma_{\beta\alpha}^m) (\gamma_m)_\gamma \delta \partial \theta^\gamma \partial \theta^\delta + \frac{1}{8} (\gamma_{\beta\epsilon}^m) (\gamma_m)_\gamma \delta \theta^\epsilon \partial \theta^\gamma \partial \theta^\delta$$

$$+ (\gamma_{\alpha\epsilon}^m) (\gamma_m)_\gamma \delta \theta^\epsilon \partial \theta^\gamma \frac{d}{d\tau}) \delta(\sigma - \sigma')$$

$$= (-\frac{1}{8} (\gamma_{\alpha\beta}^m) (\theta \gamma_m \dot{\theta}) + \frac{1}{8} (\gamma_{\beta\alpha}^m) (\gamma_m \theta)_\beta (\gamma_m \theta)_\alpha$$

$$+ \frac{1}{8} (\gamma_{\beta\epsilon}^m) (\gamma_m \theta)_\beta (\gamma_m \theta)_\alpha \frac{d}{d\tau}) \delta(\sigma - \sigma')$$

$$\therefore -\gamma_{\alpha\beta}^m \Pi_m \delta(\sigma - \sigma') - \frac{1}{2} (\gamma_{\alpha\beta}^m) (\theta \gamma_m \dot{\theta}) \delta(\sigma - \sigma')$$

$$= -\gamma_{\alpha\beta}^m (\Pi_m + \frac{1}{2} (\theta \gamma_m \dot{\theta})) \delta(\sigma - \sigma') + \dots$$

$$= -\gamma_{\alpha\beta}^m \bar{\Pi}_m + \dots$$

~ other terms >

$$[\frac{1}{4} (\gamma_{\beta\alpha}^m) (\gamma_m \dot{\theta})_\beta (\gamma_m \dot{\theta})_\alpha + \frac{1}{4} (\gamma_{\beta\alpha}^m) (\gamma_m \dot{\theta})_\beta$$

$$+ ((\gamma_{\beta\alpha}^m) (\gamma_m \dot{\theta})_\alpha (\gamma_m \dot{\theta})_\beta + ((\gamma_{\beta\alpha}^m) (\gamma_m \dot{\theta})_\alpha (\gamma_m \dot{\theta})_\beta) \frac{d}{d\tau}] \delta(\sigma - \sigma')$$

$$\rightarrow \left(\frac{1}{2} (\lambda \gamma^\mu \theta) (\lambda \gamma_\mu \bar{\theta}) + \frac{1}{2} (\lambda \gamma^\mu \theta) (\lambda \gamma_\mu \bar{\theta}) \frac{d}{d\sigma} \right) \delta(\sigma - \sigma')$$

Use Fierz so that

$$(\lambda \gamma^\mu \theta) (\lambda \gamma_\mu \bar{\theta}) \propto (\lambda \gamma^\mu \lambda) (\theta \gamma_\mu \bar{\theta}) = 0$$

This leaves

$$\{d\alpha, d\beta\} = -\pi_{\alpha\beta} \quad \text{and so finally:}$$

$$\{Q, Q\} = Q^2 = \oint \lambda^\alpha \lambda^\beta (-\gamma_{\alpha\beta}^\mu \pi_\mu)$$

$$\bullet \delta d\alpha = \{d\alpha, Q\} = -(\pi \lambda)_\alpha$$

* For CFT: OPE $\leftrightarrow$ Poisson Structure

$$\text{ex: } \{\theta^\alpha(\sigma), p_\beta(\sigma')\} = \delta^\alpha_\beta \delta(\sigma - \sigma')$$

$$\left[\begin{array}{l} p_\alpha(z) \theta^\beta(w) \sim \frac{\delta^\beta_\alpha}{z-w} \end{array} \right.$$

$$\delta(x) \rightarrow \frac{1}{z-w}$$

$$\text{so for } \{d\alpha, d\beta\} = \pi_{\alpha\beta} \delta(\sigma - \sigma')$$

$$\rightarrow d\alpha(z) d\beta(w) \sim \frac{\pi_{\alpha\beta}}{z-w}$$

After Gauge Fixing:

$$S_{ps} = \int d^2z \left(-\frac{1}{2} \partial \bar{\lambda}^\mu \bar{\partial} \lambda_\mu + \omega_\alpha \bar{\partial} \lambda^\alpha - p_\alpha \bar{\partial} \theta^\alpha \right. \\ \left. \hat{\omega}_\alpha \hat{\lambda}^\alpha - \hat{p}_\alpha \bar{\partial} \hat{\theta}^\alpha \right)$$

(Gauge fixed "minimal" Pure Spinor)

* Tree-level measure factor needs regularization:

[1] From the fact that λ^α are no longer projective and hence compact. (We have fixed the Scale Symmetry).

[2] Fixed $\bar{\lambda}$ breaks Lorentz

$\leadsto$ Can fix both by going to non-minimal: introduce new fields $\bar{\omega}$

$$S = S_{ps} + \int d^2z \left(\underbrace{\bar{\omega}^\alpha \bar{\partial} \bar{\lambda}_\alpha}_{\text{Bosons}} + \underbrace{s^\alpha \bar{\partial} r_\alpha + \bar{\omega}^\alpha \bar{\partial} \bar{\lambda}_\alpha + \hat{s}^\alpha \bar{\partial} \hat{r}_\alpha}_{\text{Fermi}} \right)$$

* In order that $(\bar{\omega}, \bar{\lambda}, \dots)$ do not contribute to the cohomology we must redefine Q

$$Q_{BRST} = Q + \int dz \tilde{w}^\alpha \Gamma_\alpha$$

$$\tilde{Q} = Q + \int dz \tilde{w}^\alpha \Gamma_\alpha$$

* Can arrive at (non-mixed) from mixed by introducing
Bosonic fields: Plus constraints

$$S_{PS}^m = \int dz d\sigma \left[P_m \dot{x}^m + w_\alpha \nabla \lambda^\alpha + \tilde{w}_\alpha \nabla \tilde{\lambda}^\alpha + \underbrace{\tilde{w}^\alpha \nabla \tilde{\lambda}_\alpha}_{\text{ghost}} + \underbrace{\tilde{w}^\alpha \nabla \tilde{\lambda}_\alpha}_{\text{ghost}} \right]$$

$$H_\alpha \tilde{w}^\alpha + H_\alpha \tilde{w}^\alpha + \dots$$

* H : Lagrange Multipliers

* Γ^α : Ghost fields

$$\Rightarrow Q_{\text{new}} = \text{'(Constraint) } \times \text{'(ghost)'} \\ = \int dz \tilde{w}^\alpha \Gamma_\alpha$$

$$\tilde{Q} = Q + Q_{\text{new}}$$

* $\tilde{\lambda}$ are also pure Spinors: $\tilde{\lambda} \gamma^m \tilde{\lambda} = 0$

→ In order to have Γ^α be the superpartner of $\tilde{\lambda}$ and not introduce any new D.O.F. then $\tilde{\lambda} \gamma^m \tilde{\lambda} = 0$ must be invariant under

Supersymmetry

$$\left\{ \begin{array}{l} \delta \tilde{\lambda}_\alpha = \Gamma_\alpha, \quad \delta S^\alpha = \tilde{w}^\alpha \\ \delta \Gamma_\alpha = 0, \quad \delta \tilde{w}^\alpha = 0 \end{array} \right\}$$

$$\Rightarrow \delta(\tilde{\lambda} \gamma^m \tilde{\lambda}) = \tilde{\lambda}_\alpha (\gamma^m)^{\alpha\beta} \Gamma_\beta = 0$$

Tree Amplitude:

$$\mathcal{A} = \langle V_1(z_1) V_2(z_2) V_3(z_3) \prod_{i=4}^N \int dz_i V_i(z_i) e^{-(\lambda(y) \tilde{\lambda}(y) + \Gamma(y) \Theta(y))} \rangle$$

* Replace PCO with:

$$\mathcal{N}(y) \equiv e^{\{Q, -\tilde{\lambda}(y) \Theta(y)\}}$$

$$\{Q, \tilde{\lambda}(y) \Theta(y)\} = Q \tilde{\lambda}(y) \Theta(y) + \tilde{\lambda}(y) \Theta(y) Q$$

$$= Q \tilde{\lambda} \Theta(y) + \tilde{\lambda}(y) (Q \Theta - \beta)$$

$$= 0 \lambda \Theta + (-0 \lambda \Theta + \Gamma \Theta) - \tilde{\lambda} \beta$$

$$= Q\lambda\theta - Q\lambda\theta + r\theta + \bar{\lambda}B$$

$$\Rightarrow \delta\theta = \lambda$$

$$\therefore -\{Q, \bar{\lambda}\theta\} = -(\lambda\bar{\lambda} + r\theta)$$

$$A = \int [D^{10}x] [D^6\theta] [D^4\lambda] [D^4\bar{\lambda}] [D^4r] e^{-S(x, \theta, \lambda, \bar{\lambda}, r)} e^{-S(x, \theta, \lambda, \bar{\lambda}, r)}$$

Can modify the Gauge fixing condition $(\bar{\lambda}, r)$

$$\chi = p_\alpha \lambda^\alpha + \partial_\theta^\alpha \lambda_\alpha + \delta^\alpha H_\alpha - e \bar{\lambda}_\alpha \theta^\alpha$$

$$Q\chi = p_\alpha \partial_\theta^\alpha \lambda^\alpha + \delta^\alpha \bar{\lambda}_\alpha - e(\bar{\lambda}_\alpha \lambda^\alpha + r_\alpha \theta^\alpha)$$

$$Q\bar{\lambda}_\alpha = r_\alpha$$

* The $\int Q\chi \sim \{Q, \Lambda\}$ which acts like a regulator for the integration measure. This is similar to the cohomological set up for the topological string.

* $\delta A_\alpha = \nabla_\alpha \Omega$, Ω : scalar superfield

* Regulator is: $I + Q\Omega$

* Let $\Lambda \equiv -\bar{\lambda}_\alpha \theta^\alpha$, then

$$Q\theta^\alpha =$$

$$\begin{aligned} Q\lambda &= -r_\alpha \theta^\alpha - \bar{\lambda}_\alpha Q\theta^\alpha \\ &= -r_\alpha \theta^\alpha - \bar{\lambda}_\alpha \lambda^\alpha \end{aligned}$$

* $\int$

$$\int d^{10}x d^6\theta d^4\lambda d^4\bar{\lambda} d^4r e^{-x^\alpha \bar{\lambda}_\alpha - r_\alpha \theta^\alpha} f(x, \theta, \lambda)$$

regularizes integral over λ since f does not depend on r

$$= \tau^{\alpha\beta\gamma\delta} \delta_{\alpha\beta} \delta_{\gamma\delta} \int d^{10}x \int d^6\theta \left(\frac{\partial}{\partial \lambda} \right)^3 f(x, \theta, \lambda)$$

T is defined by

$$(\lambda^\alpha \theta^\beta) (\lambda^\gamma \theta^\delta) (\lambda^\epsilon \theta^\zeta) (\lambda^\eta \theta^\theta) = T_{\alpha\beta\gamma\delta\epsilon\zeta\eta\theta} \lambda^\alpha \lambda^\beta \lambda^\gamma \lambda^\delta \lambda^\epsilon \lambda^\zeta \lambda^\eta \theta^\theta$$

* Use $\bar{\lambda}^\alpha \theta^\alpha = 0$ to show

$$\int d^4 r = \tau^{\alpha\beta\gamma\delta} \delta_1 \dots \delta_5 \epsilon_{\delta_1 \dots \delta_5} \frac{\partial}{\partial \tau^{\delta_1}} \dots \frac{\partial}{\partial \tau^{\delta_5}} (\bar{\lambda}_a \bar{\lambda}_b \bar{\lambda}_c)$$

$$\int d^{10} x d^{16} \theta d^{11} \lambda d^{11} \bar{\lambda} d^4 r e^{-\bar{\lambda}^a \lambda_a - \tau_a \theta^a} f(x, \theta, \lambda)$$

$$\int d^4 r e^{-\tau_a \theta^a} = (-1)^{11} \theta_1 \dots \theta_{11}$$

$$\Rightarrow \int d^{10} x d^5 \theta d^{11} \lambda d^{11} \bar{\lambda} (-1)^{11} \theta_1 \dots \theta_{11}$$

Note:

$$d\theta_{12} d\theta_{13} \dots d\theta_{16} \theta_1 \dots \theta_{11} = c \epsilon^{\epsilon_1 \dots \epsilon_{11}} \delta_{\epsilon_1 \dots \epsilon_{11}}$$